

STAGE 2 SUBMISSION - 1-PAGE EXECUTIVE SUMMARY

Applicant: Dipta Dhor | **Track:** Track E - AI/ML: Auto-Marking

What Was Done:

Designed, implemented, and evaluated an automated grading prototype using Gemini 3.6 Flash with structured JSON output to grade GCSE/11+ short answers against official mark schemes.

Technical Approach:

1. Implemented zero-shot prompting with `temperature=0.0` for deterministic scoring.
2. Enforced structured JSON output (`response_mime_type="application/json"`) returning `score` and `feedback` fields.
3. Evaluated performance across 10 sample test cases featuring full-mark, partial-credit, incorrect, and off-topic responses.

Key Assumptions:

1. Questions target short textual answers (1–3 marks).
2. Low API latency is suitable for real-time student evaluation workflows.
3. Human-in-the-loop review queues catch edge-case phrasing variations.